Working with APIs

Python code:

```
import requests
import random
import logging
import time

# --------------------------------
# CONFIGURATION
# --------------------------------
BASE_URL = "https://vedicscriptures.github.io/slok"
MAX_RETRIES = 3

# Chapter-wise verse count (accurate)
CHAPTER_VERSE_COUNT = {
    1: 47, 2: 72, 3: 43, 4: 42, 5: 29, 6: 47,
    7: 30, 8: 28, 9: 34, 10: 42, 11: 55, 12: 20,
    13: 35, 14: 27, 15: 20, 16: 24, 17: 28, 18: 78
}

# Logging setup
logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s - %(levelname)s - %(message)s"
)

# --------------------------------
# FETCH SHLOKA FUNCTION
# --------------------------------
def fetch_shloka():
    for attempt in range(1, MAX_RETRIES + 1):
        try:
            chapter = random.randint(1, 18)
            verse = random.randint(1, CHAPTER_VERSE_COUNT[chapter])

            url = f"{BASE_URL}/{chapter}/{verse}"

            logging.info(f"Requesting Chapter {chapter}, Verse {verse} (Attempt {attempt})")

            response = requests.get(url, timeout=5)

            if response.status_code != 200:
```

```
            raise Exception(f"HTTP {response.status_code}")

        data = response.json()

        return {
            "chapter": chapter,
            "verse": verse,
            "shloka": data.get("slok", "Not available"),
            "meaning": data.get("tej", "Meaning not available")  # 'tej' = English meaning
        }

    except Exception as e:
        logging.warning(f"Attempt {attempt} failed: {e}")
        time.sleep(1)

    return {"error": "Failed after retries"}

# -------------------------------
# MAIN EXECUTION
# -------------------------------
if __name__ == "__main__":
    result = fetch_shloka()

    if "error" in result:
        print("❌ Error:", result["error"])
    else:
        print("\n📖 Bhagavad Gita Shloka")
        print(f"Chapter: {result['chapter']} | Verse: {result['verse']}")

        print("\n🕉️ Shloka:\n")
        print(result["shloka"])

        print("\n💡 Meaning:\n")
        print(result["meaning"])
```

Terminal Result:

```
C:\Users\kiran\OneDrive\Desktop\Python Code>py PyCode.py
2026-03-22 19:44:19,713 - INFO - Requesting Chapter 3, Verse 35 (Attempt 1)

📖 Bhagavad Gita Shloka
Chapter: 3 | Verse: 35

🕉️ Shloka:

श्रेयान्स्वधर्मो विगुणः परधर्मात्स्वनुष्ठितात् |
स्वधर्मे निधनं श्रेयः परधर्मो भयावहः ||३-३५||

💡 Meaning:

{'author': 'Swami Tejomayananda', 'ht': '||3.35|| सम्यक् प्रकार से अनुष्ठित परधर्म की अपेक्षा गुणरहित स्वधर्म का
पालन श्रेयष्कर है;\xa0 स्वधर्म में मरण कल्याणकारक है (किन्तु) परधर्म भय को देने वाला है।।'}
```

```
C:\Users\kiran\OneDrive\Desktop\Python Code>py PyCode.py
2026-03-22 19:44:36,603 - INFO - Requesting Chapter 18, Verse 20 (Att
empt 1)

📖 Bhagavad Gita Shloka
Chapter: 18 | Verse: 20

🕉️ Shloka:

सर्वभूतेषु येनैकं भावमव्ययमीक्षते |
अविभक्तं विभक्तेषु तज्ज्ञानं विद्धि सात्त्विकम् ||१८-२०||

💡 Meaning:

{'author': 'Swami Tejomayananda', 'ht': '||18.20|| जिस ज्ञान से मनुष्य, विभक्त रूप में स्थित समस्त भूतों में एक अविभक्त और अविनाशी (अव्यय) स्वरूप को देखता है, उस ज्ञान को तुम सात्त्विक जानो।।'}

C:\Users\kiran\OneDrive\Desktop\Python Code>
```

So its different output at each execution.

Folder structure:

```
PYTHON CODE
  Dockerfile
  PyCode.py
  requirements.txt
```

Dockerfile | requirements.txt | PyCode.py

Dockerfile > ...

```
FROM python:3.10-slim

WORKDIR /myapp

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY ./PyCode.py .

CMD [ "python", "PyCode.py" ]
```

To create a image:

```
p\Python Code> docker build .

ion from Dockerfile
21B
docker.io/library/python:latest


python:latest@sha256:0040600478
python:latest@sha256:0040600478
```

```
=> [2/5] WORKDIR /myapp
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY ./PyCode.py .
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker images -a

IMAGE                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
java-input-app:latest                  2ee3d21b2faa        636MB         198MB
mywebapp:01                            89eacf391531       2.11GB         491MB
mywebapp:02                            b90c1a14ca28       2.11GB         491MB
nginx:latest                           dec7a90bd097        240MB        65.8MB
openjdk:27-ea-trixie                   4a7da6fcd5de       1.25GB         396MB
python:latest                          0040600478fe       1.63GB         432MB
routhkiranbabu/webapp-demo:01          7023f47e1922       2.11GB         491MB
routhkiranbabu/webapp-demo:02          b90c1a14ca28       2.11GB         491MB
<untagged>                             dd2e35854495        199MB        48.5MB
<untagged>                             d209b8ea96db       1.61GB         414MB
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker run dd2e35854495
2026-03-22 14:28:16,356 - INFO - Requesting Chapter 6, Verse 38 (Attempt

📖 Bhagavad Gita Shloka
Chapter: 6 | Verse: 38

🕉Shloka:

कच्चिन्नोभयविभ्रष्टश्छिन्नाभ्रमिव नश्यति |
अप्रतिष्ठो महाबाहो विमूढो ब्रह्मणः पथि ||६-३८||

💡 Meaning:

{'author': 'Swami Tejomayananda', 'ht': '||6.38|| हे महाबाहो! क्या वह ब्रह्म
ता है?'}
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```